

RFQ2BOQ — Internship Final Review

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Slide 1 — Title

RFQ2BOQ Automated Bill of Quantities Extraction from Construction Tender Documents using NLP

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Slide 2 — The Problem

- Indian construction tenders (RFQs) arrive as **50–100+ page** unstructured PDFs/Excels
- Estimators manually locate BOQ tables buried among eligibility clauses, specs, annexures
- Manual transcription is **slow** and **error-prone** — a missed row has direct commercial impact
- No tolerance for data loss in a business context

Slide 3 — What We Built

A pipeline: PDF/Excel → structured, unpriced BOQ (Excel + JSON)

Built to 4 explicit client requirements: - **R1** — 100% data-conversion fidelity (flag uncertain rows, never drop) - **R2** — GeM portal catalog as authoritative material reference - **R3** — Real 100+ document data foundation + annotation pipeline - **R4** — Structure-first extraction for large tenders (mirrors how a human estimator navigates)

Slide 4 — Architecture

PDF/Excel → Structure-first section routing → Rule/dictionary entity extraction → GeM catalog validation → BOQ assembly + unit normalization → Fidelity audit → Excel/JSON export (flagged rows visible)

- Structure-first: scans document outline first, routes extraction only to BOQ-likely sections
- Never processes a 100-page tender page-by-page blindly

Slide 5 — The Fidelity Guarantee (R1) — Proven, Not Just Claimed

A **dedicated fidelity-audit tool** compares pipeline output against an independent, source-document-derived row count — never against the pipeline's own output.

Reference set result (reproduced live):

Metric	Value
Total source rows	193
Captured	163
Flagged (surfaced for review)	34
Dropped	0
Overall fidelity	102.1% PASS

Every row is either captured with confidence, or explicitly flagged — never both invisible and gone.

Slide 6 — GeM Catalog Validation (R2)

- Client's own submitted GeM product list ingested **verbatim**, with provenance tracking
- 19 products, byte-exact from the client's source file
- Any material extracted from a GeM tender that doesn't match the catalog → **flagged**, not silently accepted
- Rationale: GeM buyers/sellers operate from the same standardized list — a non-match signals a likely extraction error

Slide 7 — Data Foundation (R3)

- **127 real tender documents** collected, hashed for integrity, organized
- Frozen train/dev/test split — test set never used for training, keeps final evaluation honest
- Annotation-review tool built: draft candidates → human review → verified training data
- **32,933 candidate sentences** already drafted and queued, ready for review

Slide 8 — Challenges Faced (the real story)

- 1. Root-caused a hidden data problem** Inherited model scored ~99% on its own test set, ~0.43 on real tenders — traced to training data that was auto-generated by regex-matching research papers, not real annotated tenders. Circular, not real.
- 2. Verified AI-agent work — didn't trust it** Built independent re-verification into the workflow after finding multiple instances of agents editing test expectations or reference data to match broken output instead of fixing the underlying bug.
- 3. Caught a real regression before it shipped** A new extraction path was silently preferred over a reliable one whenever it produced *any* output — dropped one document's fidelity from 100% to 46.7%. Root-caused and fixed with an explicit quality-comparison check, verified across the full reference set.

Slide 9 — Results Summary

Requirement	Status
R1 — Fidelity engineering	■ Complete, proven, reproducible
R2 — GeM catalog	■ Complete
R3 — Data foundation	■ Complete (annotation tooling ready)
R4 — Structure-first extraction	■ Complete
Domain-trained NER model	■ Gated on human-reviewed data (in progress)

Slide 10 — Honest Assessment

- Current entity extraction is **rule-based** (regex + dictionary) — a real, tested, usable baseline
- **Not yet** a domain-trained neural model — that step requires human-annotated data
- This matches the project's own source literature: *"data volume, not architecture, is the lever for NER accuracy"*
- Reported honestly rather than overstated — engineering integrity was itself a core deliverable of this internship

Slide 11 — Learning Outcomes

- Systematic debugging & root-cause analysis
- Data integrity / provenance system design
- Orchestrating AI-agent-driven development with independent verification
- Test-driven and regression-driven engineering discipline
- Domain knowledge: Indian tender conventions, GeM portal structure
- Professional, honest technical communication under deadline pressure

Slide 12 — Future Scope

1. Complete human review of queued annotation candidates (1,000+ sentence target)
2. Train and integrate a domain-specific NER model
3. Extend 100%-fidelity guarantee across the full 127-document corpus
4. Controlled, honest final evaluation against the frozen test set
5. Continuous intake — every new tender SWA receives gets the same fidelity treatment automatically

Slide 13 — Conclusion

- Delivered a working, fidelity-audited extraction pipeline meeting the client's explicit engineering requirements
- Built and demonstrated a verification discipline that caught real integrity failures during development
- Remaining core work (trained NER model) is honestly reported, correctly scoped, and ready to proceed
- Strong foundation for SWA to continue building on

Thank you.